Neurology & Psychiatry OSCE Cases

Question and answer review by exam year, 2018 to 2024

Key: Drug names **Critical point** ★ Repeated case Answers follow AHA/ASA stroke, ACLS, ADA and APA guidance.

2024

Case 1: Acute Stroke (tPA Eligibility, Bradycardia, Aspiration Pneumonia) ★ Repeated case, also in 2018–2019

Case details: Stroke symptoms with sinus bradycardia and hypotension, and aspiration pneumonia. **Spinal cord injury 6 weeks ago.**

Question	Answer
Q1. Is the patient a candidate for tPA?	• **No:** spinal cord injury/spinal surgery or severe trauma within the previous 3 months is a contraindication • Other exclusions to check: ICH on CT, BP >185/110 not controlled, platelets <100,000, INR >1.7, recent major surgery or GI bleed, on a DOAC within 48 h, glucose <50 • Alternative: mechanical thrombectomy if a large-vessel occlusion within 24 hours; Aspirin 160–325 mg within 24–48 h (after a dysphagia screen); BP permissive up to 220/120
Q2. Atropine 0.5 mg up to 3 mg given for bradycardia and hypotension; still not responding. Next step?	• Atropine dosing (ACLS): 1 mg IV q3–5 min, maximum 3 mg • Next: **Dopamine infusion 5–20 mcg/kg/min** (or Epinephrine 2–10 mcg/min), or transcutaneous pacing • Spinal cord injury → neurogenic shock: Norepinephrine is preferred to keep MAP 85–90 mm Hg for up to 7 days • Monitor: ECG, BP, urine output, neurologic status
Q3. What are the risk factors for aspiration pneumonia?	• In this patient: stroke (dysphagia, reduced consciousness) and spinal cord injury (weak cough, immobility) • Others: supine position, mechanical ventilation, enteral feeding tube, GERD, seizures, sedatives, esophageal stricture or fistula, COPD • Prevention: NPO until a swallow screen, head of bed 30–45°, oral care • Treatment if it occurs: Ampicillin-sulbactam 3 g IV q6h (anaerobic coverage only for abscess or empyema) • Non-pharm: NPO until swallow screen, speech-language therapy, head of bed 30–45°, oral care, upright for meals • Monitor: temperature, RR, SpO2, WBC, chest X-ray, sputum culture, swallow reassessment at 24–72 h, clinical response in 48–72 h
Also be ready for (exam add-ons)	• Goals: restore perfusion, limit infarct size, prevent complications (aspiration, VTE), secondary prevention • Non-pharm: stroke unit care, swallow screen, early mobilization and rehabilitation, glucose 140–180 mg/dL, head of bed, BP per protocol • First-line: Alteplase 0.9 mg/kg (max 90 mg) within 4.5 h if eligible; thrombectomy for large-vessel occlusion; otherwise Aspirin 160–325 mg within 24–48 h • Pregnancy: thrombolysis can be considered case by case; Aspirin 81 mg acceptable; statins and Warfarin avoided (LMWH instead); BP: Labetalol or Nifedipine, avoid ACE inhibitors/ARBs • Renal/dialysis: Alteplase no change; DOAC dose by CrCl; contrast risk with CT angiography • Liver: INR >1.7 contraindicates Alteplase; statin caution in active liver disease • HF/AF: anticoagulate after stroke (timing by infarct size), CHA2DS2-VASc • Counseling: FAST signs, BP and glucose control, smoking cessation, adherence to antiplatelet and statin

2022–2023

Case 1: Ischemic Stroke with DM, Peripheral Neuropathy and Diabetic Foot Infection

Case details: Man with HTN and DM, symptoms of ischemic stroke; recent appendectomy (highlighted in red). BP about 150/100, HbA1c 9.5%. Smoker, obese, on Aspirin.

Question	Answer
Q1. What are the patient's risk factors?	• Male sex, smoking, obesity, hypertension, diabetes (also uncontrolled glycemia, age)
Q2. Will you give Alteplase?	• **Information is not enough:** the time of symptom onset (last known well) is missing • Also check: CT head, glucose, BP <185/110, platelets, INR, **recent appendectomy date** (major surgery within 14 days is a relative contraindication: weigh risk and benefit)
Q3. They decided to give Alteplase. What is the dose?	• 0.9 mg/kg (maximum 90 mg): 10% as an IV bolus over 1 minute, the remaining 90% as infusion over 60 minutes • Alternative: Tenecteplase 0.25 mg/kg single bolus (maximum 25 mg) • Monitor: neuro checks and BP every 15 min × 2 h, then every 30 min × 6 h, then hourly to 24 h; stop infusion for severe headache, vomiting, or neurological decline
Q4. What is the time window?	• Within **4.5 hours** of last known well (3 hours for the original labeling) • Beyond that: imaging-selected patients or thrombectomy up to 24 hours

Question	Answer
Q5. What will you do with Aspirin? For how long?	• **Hold antiplatelets for 24 hours after tPA** (and until the follow-up CT excludes hemorrhage); then restart • Blood pressure before tPA <185/110 and <180/105 for the first 24 h (Labetalol or Nicardipine)
Q6. What medications are given other than Alteplase?	• Antiplatelet therapy; DAPT for minor stroke or high-risk TIA • High-intensity statin: Atorvastatin 80 mg (LDL <70 mg/dL) • Anticoagulant (timing individualized) if cardioembolic stroke (AF) • Glucose 140–180 mg/dL; VTE prophylaxis (IPC); swallow screen; BP management • Non-pharm: stroke unit care, early mobilization and rehabilitation (PT/OT), smoking cessation, diabetic and BP control, dysphagia screen, VTE prevention (compression devices) • Monitor: NIHSS and neuro checks, BP (<180/105 for 24 h after tPA), glucose, lipids and LFTs/CK on statin, bleeding signs, platelets, CT head at 24 h
Q7. Which DAPT will you give?	• Clopidogrel 75 mg daily (300 mg load) + Aspirin 81 mg daily (162–325 mg load), started within 24 hours
Q8. For how long? Why this duration for Clopidogrel?	• **Minor stroke (NIHSS ≤3) or high-risk TIA: 21 days** (up to 90 days in some trials), then a single antiplatelet indefinitely • Benefit is largest in the first 21 days; bleeding risk grows with longer DAPT • Major stroke: single antiplatelet only • Monitor: bleeding (GI, intracranial), Hgb, adherence, repeat neurologic exam, stop DAPT at day 21
Q9. What is the A1C target? Why and which factors?	• **<7%** for most adults • Factors: comorbidities, life expectancy, disease duration, established vascular complications, hypoglycemia risk, patient preference
Q10. Peripheral neuropathy with new depression: which agent?	• **Duloxetine** 30 mg daily × 1 week, then 60 mg daily (SNRI treats both)
Q11. What are the monitoring parameters for Duloxetine?	• BP and HR, renal function (avoid CrCl <30), LFTs (avoid in hepatic disease/alcohol), sodium (SIADH), blood glucose and A1C • Behavior change and suicidality, serotonin syndrome, bleeding with Aspirin; QTc if other QT drugs
Q12. MRSA foot infection on Linezolid; current drugs: Lisinopril, Linezolid, Aspirin, Duloxetine, Metformin. Major interactions and risks?	• **Linezolid + Duloxetine:** Linezolid is a weak MAO inhibitor → **serotonin syndrome** (avoid; if unavoidable stop Duloxetine and monitor) • **Linezolid + Aspirin:** Linezolid causes thrombocytopenia (especially beyond 10–14 days) → bleeding risk • Linezolid + Metformin/antidiabetics: hypoglycemia and lactic acidosis • Safer oral alternatives for MRSA: Doxycycline 100 mg BID or TMP-SMX DS BID (mild–moderate infection); a single long-acting IV (Dalbavancin) if admission is refused
Also be ready for (exam add-ons)	• Goals: reperfusion, prevent recurrence (BP, LDL <70, A1C <7%), treat neuropathy and depression, treat the infection • Non-pharm: stroke unit, rehabilitation, smoking cessation, weight loss, diet, foot care • First-line: Alteplase 0.9 mg/kg; Aspirin 81 mg + Clopidogrel 75 mg × 21 days (minor stroke); Atorvastatin 80 mg; Duloxetine 30 → 60 mg • Pregnancy: statins, ACE inhibitors/ARBs, Duloxetine and Metformin (limited) require review; use Labetalol and Insulin • Renal/dialysis: Duloxetine avoid CrCl <30; Metformin stop eGFR <30; Linezolid no adjustment • Liver: Duloxetine avoid in hepatic impairment/alcohol; statin caution • HF: avoid Pioglitazone; Counseling: Duloxetine: nausea, sweating, BP; do not stop abruptly; Linezolid + SSRI/SNRI = serotonin syndrome; foot inspection daily

2021

Case 1: Focal Seizures with Carbamazepine in a Patient on Warfarin ★ Repeated case, also in 2018–2019

Case details: Antiphospholipid syndrome with recurrent DVT and AFib on Warfarin. New partial focal seizures. The neurologist wants Carbamazepine, is insistent and

does not want to change the plan.

Question	Answer
Q1. What is the interaction, what do you recommend, and how do you handle the neurologist?	• **Interaction:** Carbamazepine induces CYP3A4, CYP2C9 and CYP1A2 → lower Warfarin levels and **INR falls**; effect builds over 1–4 weeks and lasts weeks after stopping; Carbamazepine also induces its own metabolism • Risk: sub-therapeutic INR → recurrent thrombosis (antiphospholipid syndrome is high-risk; DOACs are not suitable) • **Recommend a non-inducing antiseizure drug:** Levetiracetam 500 mg BID (titrate to 1000–1500 mg BID), or Lacosamide or Lamotrigine • Keeping the same Warfarin dose is not safe • If Carbamazepine is still chosen: expect a Warfarin increase (often 50–100%), INR twice weekly then weekly for 4–6 weeks, and again when it is stopped; bridge with Enoxaparin if INR <2 • **Communication (SBAR):** stay calm and respectful; acknowledge the aim of seizure control; state the facts (major interaction, thrombosis risk); offer a concrete alternative rather than only a problem; agree a monitoring plan; if disagreement continues, escalate to a senior pharmacist/consultant and document • Monitor: INR, seizure frequency, sodium (Carbamazepine), CBC, LFTs, HLA-B*1502 where relevant, drug levels
Also be ready for (exam add-ons)	• Goals: seizure control with the fewest adverse effects and without losing anticoagulation • Non-pharm: sleep, avoid triggers (alcohol, sleep deprivation, flashing lights), driving restrictions, seizure first aid, medical alert • First-line: focal seizures: Lamotrigine (25 mg daily, slow titration), Levetiracetam 500 mg BID, Lacosamide or Carbamazepine 200 mg BID • Pregnancy: Lamotrigine and Levetiracetam preferred; avoid Valproate; give Folic acid 1–4 mg; **Warfarin is teratogenic (6–12 weeks): switch to LMWH (APS: LMWH + Aspirin)**; enzyme-inducing drugs reduce oral-contraceptive efficacy (IUD or progestin) • Renal/dialysis: Levetiracetam adjust (CrCl <50) with a supplemental dose after HD; Lacosamide max 300 mg/day if CrCl <30 • Liver: Lamotrigine reduce dose 25–50% (moderate–severe); avoid Valproate; Carbamazepine hepatotoxic • HF: Carbamazepine can cause AV block (ECG); Counseling: Carbamazepine: dizziness, diplopia, rash/SJS (HLA-B*1502), hyponatremia, take with food; Lamotrigine: rash (slow titration); never stop abruptly; INR monitoring

Case 2: Depression and Sleep Problems (Reference Topic, No Case Details) ★ Topic listed for 2020

Case details: Reference list (2020): Psychiatry topics were depression and sleep problems. No case details were given: below is a generic approach to state your assumptions.

Question	Answer
Q1. How do you assess and treat depression?	• Assess: PHQ-9, suicide risk, bipolar screen, medical causes (thyroid, B12), substances, drug causes • First line: SSRI (Sertraline 50 mg daily, titrate to 100–200 mg; or Escitalopram 10 mg); psychotherapy (CBT) • Choose by comorbidity: poor sleep and appetite → Mirtazapine 15 mg at bedtime; avoid SSRIs with Warfarin/NSAIDs/antiplatelets if possible (bleeding) • Monitor: response at 2–4 weeks; full trial 6–8 weeks; suicidality in the first weeks; sodium in older adults • Duration: continue 6–12 months after remission (longer if recurrent) • Non-pharm: psychotherapy (CBT), exercise, sleep hygiene, social support, alcohol avoidance
Q2. How do you treat insomnia?	• First line: **CBT-I** and sleep hygiene (fixed schedule, no screens, limit caffeine/alcohol, bed for sleep only) • Short term: Zolpidem 5 mg at bedtime (women and elderly), 5–10 mg (men), low-dose Doxepin 3–6 mg, or an orexin antagonist (Suvorexant, Lemborexant) • Avoid benzodiazepines and Z-drugs in older adults (Beers criteria); avoid sedatives with opioids and in COPD/sleep apnea • Screen for sleep apnea and drug causes; reassess in 2–4 weeks • Monitor: sleep diary and ISI at 2–4 weeks, daytime sedation, falls, tolerance and dependence; reassess the need for medication every 4 weeks
Also be ready for (exam add-ons)	• Goals: remission of symptoms, restore function, prevent relapse; treat insomnia without dependence • Non-pharm: CBT, exercise, sleep hygiene, CBT-I, social support, avoid alcohol and caffeine • First-line: Sertraline 50 mg daily (titrate to 100–200 mg); insomnia: CBT-I first, then Zolpidem 5 mg or low-dose Doxepin • Pregnancy: Sertraline preferred; avoid Paroxetine; insomnia: CBT-I, Doxylamine-pyridoxine for nausea/sleep; avoid benzodiazepines and Z-drugs • Renal/dialysis: Sertraline no change; Mirtazapine reduce; Zolpidem no change • Liver: Sertraline reduce dose 50%; Zolpidem 5 mg maximum • HF: Sertraline is safe after MI; avoid TCAs; Citalopram max 20 mg if age >60 (QT) • Counseling: SSRI effect in 2–4 weeks, GI upset, sexual dysfunction, bleeding risk with Warfarin and NSAIDs, discontinuation syndrome, suicidality warning (<25 years), serotonin syndrome; Zolpidem: take right before bed with 7–8 h left, complex sleep behaviors